

SPEARMAN'S RANK CORRELATION COEFFICIENT

Rank correlation coefficient is useful for finding correlation between any two qualitative characteristics such as Beauty, Honesty, and Intelligence etc., which cannot be measured quantitatively but can be arranged serially in order of merit or proficiency possessing the two characteristics.

Rank correlation coefficient The rank correlation coefficient is defined as

$$r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)} \quad \text{where}$$

$\sum D^2$ = sum of squares of difference between the pairs of ranks

n = number of pairs of observation.

The value of r lies between -1 and $+1$. If $r = +1$, there is complete agreement in order of ranks and the direction of ranks is also same. If $r = -1$, then there is complete disagreement in order of ranks and they are in opposite directions.

Problem: In a quantitative aptitude test, two judges rank the ten competitors in the following order.

Competitor	1	2	3	4	5	6	7	8	9	10
Ranking of judge I	4	5	2	7	8	1	6	9	3	10
Ranking of judge II	8	3	9	10	6	7	2	5	1	4

Is there any concordance between two judges?

Soln:- R_x : Ranking by judge I and
 R_y : Ranking by judge II

R_x	R_y	$D = R_x - R_y$	D^2
4	8	-4	16
5	3	2	4
2	9	-7	49
7	10	-3	9
8	6	2	4
1	7	-6	36
6	2	4	16
9	5	4	16
3	1	2	4
10	4	6	36

$$\sum D^2 = 190$$

Here $n=10$;

$$\therefore r = 1 - \frac{6 \sum D^2}{n(n^2-1)} = 1 - \frac{6 \times 190}{990} = -0.1515$$

There is a low degree of negative rank correlation between the two judges.

Pb:- Twelve recruits were subjected to selection test to ascertain their suitability for certain course of training. At the end of the training they were given a proficiency test. The marks scored by the recruits are recorded below:

Recruit	1	2	3	4	5	6	7	8	9	10	11	12
Selection Test score (X)	44	49	52	54	47	76	65	60	62	58	50	67
Proficiency (Y) Test score	48	55	45	60	43	80	58	50	77	46	47	65

Find the rank correlation coefficient and comment on your result.

Soln:- R_x : rank of recruit selection test score.
 R_y : rank of proficiency selection test score.

X	Y	R _x	R _y	D = R _x - R _y	D ²
44	48	12	8	4	16
49	55	10	6	4	16
52	45	8	11	-3	9
54	60	7	4	3	9
47	43	11	12	-1	1
76	80	1	1	0	0
65	58	3	5	-2	4
60	50	5	7	-2	4
63	77	4	2	2	4
58	46	6	10	-4	16
50	47	9	9	0	0
67	65	2	3	-1	1

$$\sum D^2 = 80$$

Here $n = 12$

$$r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)} = 1 - \frac{6 \times 80}{12 \times 143}$$

$$= 0.7203$$

There is a high degree of positive rank correlation between the scores of selection and proficiency test.

Computation for tied observations: There may be two or more items having equal values. In such case the same rank is to be given. The ranking is said to be **tied**. In such circumstances an average rank is to be given to each individual item. For example if the value so is repeated twice at the 5th rank, the common rank to be assigned to each item is $\frac{5+6}{2} = 5.5$, which is the average of 5 and 6 given as 5.5, appeared twice.

If the ranks are tied, it is required to apply a correction factor which is $\frac{m(m^2-1)}{12}$, where m is the number of items whose ranks are common and should be repeated as many times as there are tied observations.

A slightly different formula is used when there is more than one item having the same value. The formula is

$$r = 1 - \frac{6 [\sum D^2 + C.F]}{n(n^2 - 1)},$$

where $C.F = \sum_{i=1}^k c.f_i$. Now, $c.f_i = \frac{m_i(m_i^2-1)}{12}$, $i = 1, 2, 3, \dots, k$, where m_i is the numbers of times repeated the tied rank and k is the number of distinct tied ranks occurs in both X and Y .

Pb:- Find the rank correlation coefficient for the following

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

Soln:-

X	Y	R_x	R_y	$D = R_x - R_y$	D^2
68	62	4	5	-1	1
64	58	6	7	-1	1
75	68	2.5	3.5	-1	1
50	45	9	10	-1	1
64	81	6	1	5	25
80	60	1	6	-5	25
75	68	2.5	3.5	-1	1
40	48	10	9	-1	1
55	50	8	8	0	0
64	70	6	2	4	16
					$\Sigma D^2 = 72$

$$C.F = \sum_{i=1}^k C.F_i \quad \text{where } C.F_i = \frac{m_i(m_i^2 - 1)}{12} \quad i = 1 \dots k$$

Here $k=3$, because we have 3 tied ranks (2.5, 6, 3.5) occurs in both X and Y.

In X series the tied rank 2.5 repeated 2 times.
So, $m_1 = 2$ and $C.F_1 = \frac{2(2^2 - 1)}{12} = \frac{1}{2}$

In X series the tied rank 6 repeated 3 times
So, $m_2 = 3$, and $C.F_2 = \frac{3(3^2 - 1)}{12} = 2$

In Y series the tied rank 3.5 repeated 2 times.
So, $m_3 = 2$, and $C.F_3 = \frac{2(2^2 - 1)}{12} = \frac{1}{2}$

Hence $C.F = C.F_1 + C.F_2 + C.F_3 = 3$.

Rank correlation coefficient $r = 1 - \frac{6[\sum D^2 + C.F]}{n(n^2 - 1)}$

Here $n = 10$

$$\therefore r = 1 - \frac{6[12 + 3]}{990} =$$

$$r = 0.5454$$